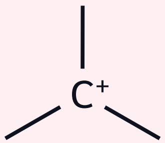
GENERAL ORGANIC CHEMISTRY — PART 2

REACTIVE INTERMEDIATES · ACIDITY & BASICITY · MECHANISM · REACTION TYPES

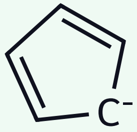
1 CARBOCATION

sp^2 planar · stability order · Bredt · 1,2-shifts & ring expansion



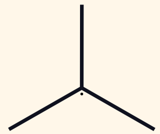
2 CARBANION

sp^3 pyramidal · reverse alkyl order · s-character · aromatic anions



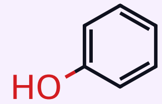
3 RADICAL & CARBENE

chain mechanism · BDE · selectivity · singlet vs triplet



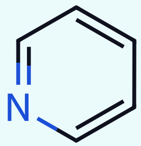
4 ACIDITY

Brønsted & Lewis · 8 factors · full pK_a ladder · ortho effect



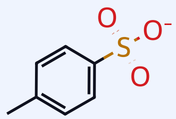
5 BASICITY

pK_b trends · aqueous vs gas phase · aromatic & heterocyclic amines



6 E⁺ · Nu⁻ · LG

who attacks whom · leaving-group ability · solvent control



7 ENERGY PROFILE

TS vs intermediate · E_a · RDS · Hammond · catalysis

$$k = A \cdot e^{-Ea/RT}$$

8 REACTION TYPES

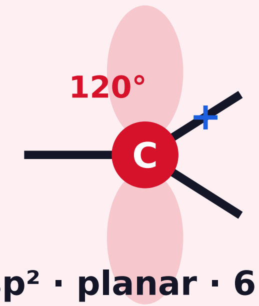
addition · substitution · elimination · rearrangement · redox

S_N1 S_N2 E1 E2

THE GOLDEN THREAD: every question here reduces to **“which species is more stable, and why?”** ★ **= MUST-MEMORISE ORDER** — learn every starred box by heart.

1 GEOMETRY DECIDES REACTIVITY

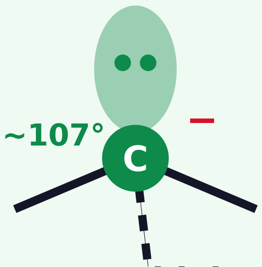
CARBOCATION



sp^2 · planar · 6 e^-

empty p · ELECTROPHILE

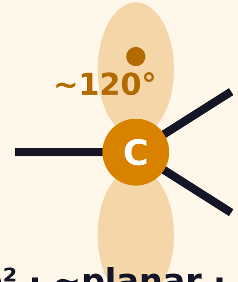
CARBANION



sp^3 · pyramidal · 8 e^-

lone pair · NUCLEOPHILE

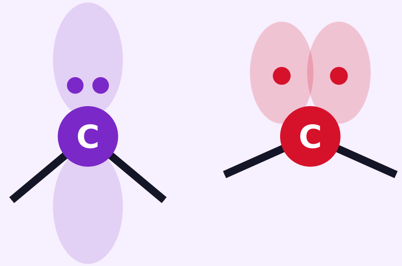
FREE RADICAL



sp^2 · ~planar · 7 e^-

1 odd e^- · PARAMAGNETIC

CARBENE



SINGLET 102° TRIPLET 136°

neutral · divalent · 6 e^-

SIDE-BY-SIDE COMPARISON

	Carbocation	Carbanion	Free radical	Carbene
Valence e^-	**6** (sextet)	**8** (octet)	**7**	**6**
Charge	+1	−1	0	0
Hybridisation	sp^2	sp^3 (sp^2 if conj.)	sp^2	sp^2
Shape	planar	pyramidal	~planar	bent
Magnetism	diamagnetic	diamagnetic	**paramagnetic**	S: dia · T: **para**
Behaves as	electrophile	nucleophile	both	electrophile
Alkyl order	3° > 2° > 1°	**1° > 2° > 3°**	3° > 2° > 1°	—
Rearranges?	**YES**	no	rare (aryl/H)	yes (to alkene)

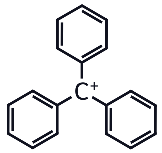
THE ONE RULE THAT UNIFIES THEM

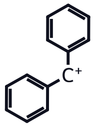
Delocalise the charge / odd electron ⟹ STABILITY ↑

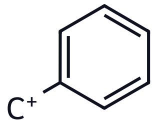
Species is…	Stabilised by
electron POOR (C^+, carbene)	+I alkyl, hyperconjugation, **+M donor**, aromaticity
electron RICH (C^-)	−I, **−M acceptor**, ↑ s-character, aromaticity
odd e^- (radical)	resonance > hyperconjugation

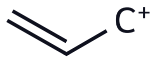
Aromaticity trumps all: $C_7H_7^+$ and $C_5H_5^-$ are the most stable simple ions

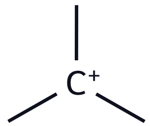
2 STABILITY ORDER — MEMORISE LEFT TO RIGHT

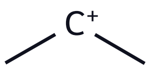

Ph_3C^+
trityl


Ph_2CH^+


$PhCH_2^+$
benzyl


allyl⁺


3°

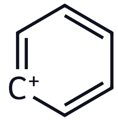

2°


1°

CH_3^+

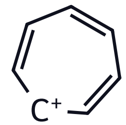
CH_3^+

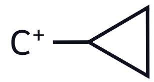

vinyl⁺

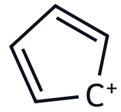

phenyl⁺

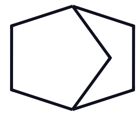
vinyl⁺ & phenyl⁺ : ⊕ on a high-s carbon ⇒ **never formed**

THE FOUR SPECIAL CASES


tropylium⁺
aromatic 6π
isolable salt


cyclopropylmethyl⁺
very stable
bent-bond overlap


cyclopentadienyl⁺
antiaromatic 4π
almost never forms


bridgehead
Bredt's rule
cannot be planar

★ **C⁺ STABILITY** **$Ph_3C^+ > Ph_2CH^+ > PhCH_2^+ \approx$ allyl⁺ > 3° > 2° > 1° > CH_3^+ > vinyl⁺ > phenyl⁺**

tropylium (aromatic 6π) and cyclopropylmethyl outrank them all · cyclopentadienyl⁺ is **antiaromatic** and almost never forms

GENERATION — 5 ROUTES

- **R-X ionises** (Ag^+, $AlCl_3$)
- **Alkene + H⁺** → Markovnikov cation
- **ROH + H⁺** → $-H_2O$ (bad LG → good LG)
- **$R-N_2^+$** → R^+ + N_2↑ (best LG)
- **Epoxide / C=O** protonated at O

FOUR THINGS A CARBOCATION CAN DO

- ① Capture Nu⁻ → S_N1 product
- ② Lose β-H → **alkene** (E1)
- ③ **Rearrange** by a 1,2-shift
- ④ Attack arene → Friedel–Crafts

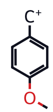
Unexpected product in a JEE question ⇒ suspect ③

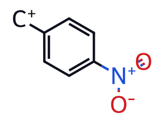
STRUCTURAL EVIDENCE

- 120° · **sp²** · empty **p** ⊥ plane
- Chiral centre destroyed ⇒ **racemisation**
- $(CH_3)_3CBr$ solvolyses ≈**10⁶×** faster than CH_3CH_2Br

TRAPS

- ✘ 3° is **not** always best — benzyl / allyl 1° beat it
- ✘ *o/p* **+M donors** ($-OCH_3$, $-NH_2$) stabilise hugely · $-NO_2$ destabilises


+M ⇒ stabilised


−M ⇒ destabilised

- ✘ Bridgehead cation in bicyclo[2.2.1] — **forbidden**

3 THE 5 LAWS OF REARRANGEMENT

- ① Only **1,2-shifts** — adjacent atom only
- ② Group moves **with its e⁻ pair**
- ③ Must be **anti-periplanar** to the empty p
- ④ Only if the new cation is **more stable**
- ⑤ Migrating group keeps its **configuration**

THE THREE SHIFT TYPES

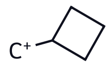
① 1,2-HYDRIDE SHIFT · 1° ⟶ 3° · commonest

$$CH_3\text{–}CH(CH_3)\text{–}CH_2^+ \rightarrow CH_3\text{–}C^+(CH_3)\text{–}CH_3$$

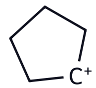
② 1,2-METHYL SHIFT · neopentyl : no S_N2

$$(CH_3)_3C\text{–}CH_2^+ \rightarrow (CH_3)_2C^+\text{–}CH_2CH_3$$

③ RING EXPANSION

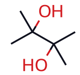


cyclobutylmethyl⁺
1° + strain



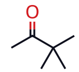
cyclopentyl⁺
2° + strain relieved

PINACOL – PINACOLONE (the exam favourite)



pinacol

H^+ ↓ $-H_2O$ ↓ 1,2-CH_3 shift



pinacolone

Driving force: 3° C^+ ⟶ **oxocarbenium**, full octet

★ MIGRATION

Ar > H > 3° R > 2° R > 1° R > CH_3

e⁻-rich aryl migrates best : *p*-CH_3O > *p*-CH_3 > C_6H_5 > *p*-NO_2 · **substrate-dependent** — in some Wagner–Meerwein systems H outruns aryl

WHEN WILL IT REARRANGE?

Mechanism	Cation formed?	Rearrangement
S_N1 / E1	**yes**	**YES — expect it**
S_N2 / E2	no	**never**
Markovnikov HX addition	**yes**	**YES**
Hydroboration–oxidation	no (4-centre TS)	**never**
Oxymercuration–demercuration	bridged Hg^+	**never**
Friedel–Crafts **alkylation**	**yes**	**YES**
Friedel–Crafts **acylation**	acylium (resonance)	**never**

PHENYL MIGRATES THROUGH A BRIDGED ION

$Ph\text{–}CH_2\text{–}CH_2^+$ **1°** $Ph\text{–}CH^+\text{–}CH_3$ **2°**

Ring π e⁻ form a **3-membered bridge** ⇒ migration is **stereospecific, retention**

60-SECOND CHECKLIST

- Draw the **first-formed** cation
- Is a **better** cation one shift away?
- Any **ring strain** next door?
- Only then write the product

★ **C⁻ STABILITY** $C_5H_5^- > Ph_3C^- > Ph_2CH^- > PhCH_2^- \approx allyl^- > HC{\equiv}C^- > CH_2{=}CH^- > CH_3^- > 1° > 2° > 3°$

aromatic ⟶ resonance ⟶ % s-character ⟶ alkyl · the alkyl part is the **exact reverse** of carbocations

4 STABILITY — FOUR INDEPENDENT LEVERS

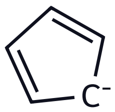
① % s-character

$HC{\equiv}C^-$ **sp**
pK_a 25

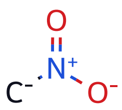
$CH_2{=}CH^-$ **sp²**
44

$CH_3CH_2^-$ **sp³**
50

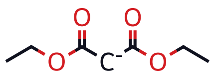
② Resonance (−M) — charge onto O / N



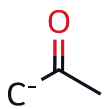
$C_5H_5^-$ **aromatic**
16



nitronate
10.2



malonate
13

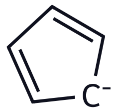


enolate
19.2

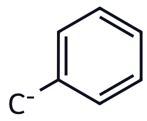
③ **−I groups** stabilise · ④ **Alkyl (+I) destabilises**

★ **$CH_3^- > 1° > 2° > 3°$**

OVERALL ORDER



cyclopentadienide



benzyl⁻

CH_3^-

CH_3^-

Shortcut: lower pK_a of the parent C–H ⇒ **more stable** carbanion

GENERATION

- **Deprotonation** — $NaNH_2$, LDA, NaH
- **R-MgX / R-Li** — polar $C^{\delta-}$–$M^{\delta+}$
- **Decarboxylation** of β-keto acid
- **Michael** conjugate addition
- **Halogen-metal exchange**

REACTIONS & STEREOCHEMISTRY

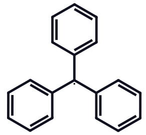
- Nu in **S_N2** & addition to **C=O**
- Aldol · Claisen · Michael · alkylation
- Intermediate of **E1cb**
- **Rapidly inverting** (like NH_3) ⇒ **racemises**
- Retains only at bridgehead / small ring

Trap: resonance-stabilised carbanions go **sp² planar** — always racemic

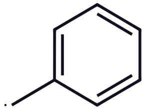
★ **R• STABILITY** $Ph_3C\bullet > Ph_2CH\bullet > PhCH_2\bullet \approx$ allyl• > 3° > 2° > 1° > $CH_3\bullet$ > vinyl• > phenyl•

same shape as the carbocation order · **lower C-H bond-dissociation energy ⟹ more stable radical**

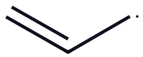
5 STABILITY & BOND-DISSOCIATION ENERGY



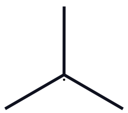
$Ph_3C\bullet$
isolable!



benzyl•



allyl•



3°

$CH_3^{\cdot}$

$CH_3\bullet$

C-H bond	BDE kJ mol⁻¹	Radical formed
CH_3–H	439	$CH_3\bullet$ (worst)
1° C–H	423	1°
2° C–H	413	2°
3° C–H	403	3°
allylic / benzylic	**371 / 375**	**best (resonance)**
vinylic / aryl	465 / 473	**never formed**

Lower BDE ⟹ more stable radical — weakest C–H attacked first

THE CHAIN MECHANISM

INITIATION (Δ or hν)
Cl–Cl ⟶ 2 Cl•

PROPAGATION (chain carriers)
Cl• + CH_4 ⟶ HCl + •CH_3
•CH_3 + Cl_2 ⟶ CH_3Cl + Cl•

TERMINATION (2 radicals meet)
Cl• + Cl• · •CH_3 + Cl• · 2 •CH_3

O_2 & hydroquinone are **inhibitors** — they kill the chain

SELECTIVITY — Cl_2 vs Br_2

Halogen	3° : 2° : 1°	Verdict
Cl• (25 °C)	**5.0 : 3.8 : 1**	fast, **unselective**
Br• (127 °C)	**1600 : 82 : 1**	slow, **very selective**

Hammond: Br• abstraction is **endothermic** ⇒ late product-like TS ⇒ selective. Cl• is exothermic ⇒ early TS ⇒ unselective.

Reactivity: $F_2 \gg Cl_2 > Br_2 \gg I_2$ (I_2 does not work — the step is endothermic)

MUST-KNOW RADICAL REACTIONS

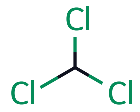
- **Kharasch:** HBr + R_2O_2 ⟶ **anti-Markovnikov**
- ✘ **HBr only** — never HCl / HF / HI
- **NBS** → allylic bromination
- Wurtz · combustion · polymerisation
- Ether **auto-oxidation** → peroxides

6 SINGLET vs TRIPLET CARBENE — THE CENTRAL COMPARISON

Property	SINGLET $:CH_2$	TRIPLET $\bullet CH_2 \bullet$
Electrons	**paired** in one sp^2 orbital	**unpaired** in 2 orbitals
Bond angle	**~102–103°**	**~136°** (wider)
Magnetism	diamagnetic	**paramagnetic**
Ground state of $:CH_2$	excited (+38 kJ)	**more stable**
Addition to alkene	**concerted, 1 step**	stepwise diradical
Stereochemistry	**STEREOSPECIFIC** *cis* ⟶ *cis* only (syn)	**NON-stereospecific** *cis* + *trans* mixture
Made in	solution / liquid phase	gas phase, sensitiser

> **Exam line:** retention ⟹ singlet · mixture ⟹ triplet. $:CCl_2$ is **singlet** (X lp → empty p).

GENERATION OF CARBENES



$CHCl_3$

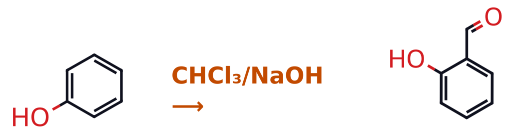
+ KOH −KCl, $-H_2O$ ⟶ **$:CCl_2$** dichlorocarbene

α-Elimination — H and Cl leave the **same** carbon

- **Diazomethane:** $CH_2N_2 \xrightarrow{h\nu} :CH_2 + N_2\uparrow$
- **Ketene:** $CH_2{=}C{=}O \xrightarrow{h\nu} :CH_2 + CO$
- **Simmons–Smith:** CH_2I_2 / Zn(Cu) → **carbenoid**, full retention

REACTIONS OF $:CCl_2$

① **Reimer–Tiemann** (phenol → salicylaldehyde)



o-hydroxy benzaldehyde

② **Carbylamine test** (1° amines only)

R–NC · foul smell

③ **Cyclopropanation** of alkenes

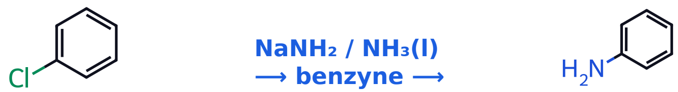
plus **C–H insertion** ($:CH_2$)

NITRENE R–N: (the nitrogen analogue — 6 e⁻, neutral, monovalent)

- **Hofmann bromamide:** $RCONH_2 + Br_2/NaOH \rightarrow RNH_2$ (**one carbon shorter**)
- **Curtius:** $RCON_3 \xrightarrow{\Delta,\ -N_2} R{-}N{=}C{=}O \rightarrow RNH_2$
- **Lossen:** from a hydroxamic acid RCONHOH
- **Schmidt:** $RCOOH + HN_3 / H_2SO_4$

> All four: **acyl nitrene** ⟶ R migrates **C → N** ⟶ isocyanate · **retention**

BENZYNE (1,2-dehydrobenzene)



chlorobenzene — aniline

- Weak **in-plane sp^2–sp^2** overlap — not a real π
- **Cine substitution** — Nu lands on either C (^{14}C proof)
- Traps as a **dienophile** (Diels–Alder)

7 DEFINITIONS

BRØNSTED-LOWRY
acid = **H^+ donor** · base = H^+ acceptor

LEWIS
acid = **e^- pair acceptor** (BF_3, $AlCl_3$, C^+) · base = donor (NH_3, R_2O)

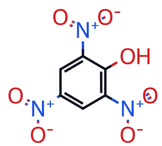
THE MASTER EQUATION
$pK_a = -\log K_a$ · $pK_a + pK_b = 14$
Lower pK_a = stronger acid. Always judge by the stability of A^-

★ ACIDIC STRENGTH RCOOH > H_2CO_3 > ArOH > H_2O > ROH > HC≡CH > NH_3 > $CH_2{=}CH_2$ > RH
pK_a 4.8 · 6.4 · 10 · 15.7 · 16–18 · 25 · 38 · 44 · 50 — always judge by the **stability of the conjugate base A^-**

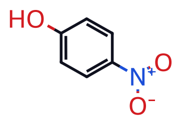
THE 8 FACTORS — IN ORDER OF POWER

- ① **Atom:** period ⇒ EN — CH_4 50 < NH_3 38 < H_2O 15.7 < HF 3.2 · group ⇒ size — HF < HCl < HBr < HI
- ② **Resonance** in A^- — biggest jump: RCOOH 4.8 ≫ ArOH 10 ≫ ROH 16–18
- ③ **Aromatic** A^- — cyclopentadiene **16**
- ④ **−I** — additive, distance-dependent
- ⑤ **Hybridisation** sp > sp^2 > sp^3
- ⑥ **Intramolecular H-bond** — salicylic 2.97 vs benzoic 4.20
- ⑦ **SIR** ⟹ acidity **falls**
- ⑧ **Solvation** of A^-

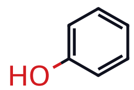
PHENOL SUBSTITUENT EFFECT (pK a)



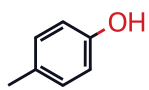
picric acid
0.38



p-nitrophenol
7.15



phenol
10.0

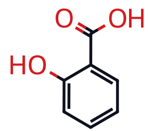


p-cresol
10.26

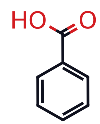
$-NO_2$ (−I,−M) ↑↑ acidity · $-CH_3$ (+I) ↓ acidity

o,p-NO_2 delocalise ⊖ onto O · *m*-NO_2 (8.4) is **−I only**

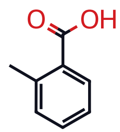
H-BOND & ORTHO EFFECT



salicylic
2.97



benzoic
4.20

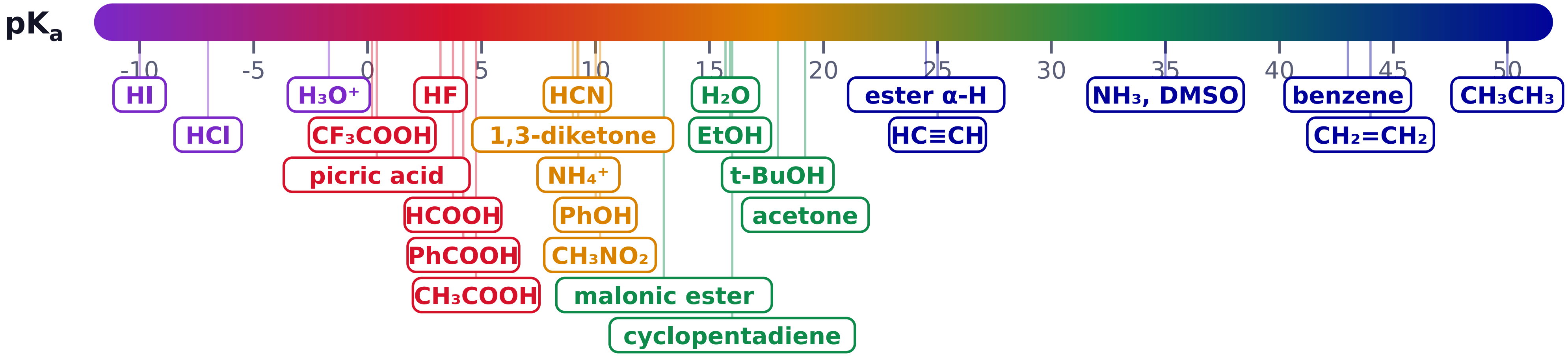


o-toluic
3.91

ORTHO EFFECT: *every ortho-*substituted benzoic acid (EDG **or** EWG) is **stronger** than benzoic acid — the −COOH is twisted out of plane

8 MEMORISE THE ANCHORS · EVERY ACID-BASE QUESTION IS SOLVED BY READING THIS LINE

STRONGER ACID ◀ low pK_a **high pK_a ▶ STRONGER BASE**

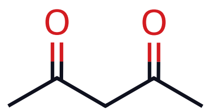


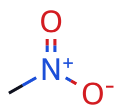
CARBOXYLIC ACIDS

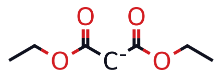
Acid	pK_a
CF_3COOH	**0.23**
Cl_3CCOOH	0.65
Oxalic (K_{a1})	1.23
$Cl_2CHCOOH$	1.29
$ClCH_2COOH$	2.86
Salicylic	2.97
HCOOH	3.75
C_6H_5COOH	4.20
CH_3COOH	**4.76**
CH_3CH_2COOH	4.87

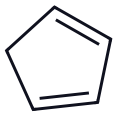
HCOOH > CH_3COOH — **no +I alkyl**

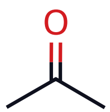
C-H ACIDS — THE α-HYDROGEN SERIES


1,3-diketone
9


CH_3NO_2
10.2


malonic ester
13


cyclopentadiene
16


acetone
19.2

★ **2 EWG > aromatic anion > 1 EWG > none**

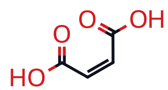
Cyclopentadiene: anion is **aromatic (6π)** ⇒ as acidic as water

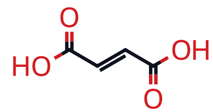
ALCOHOLS & THE GEOMETRIC ISOMER TRAP

Alcohol acidity in solution

★ **CH_3OH > 1° > 2° > 3°**

Reversed in the gas phase (no solvation)


MALEIC (*cis*)
pK_{a1} **1.92** · pK_{a2} 6.23


FUMARIC (*trans*)
pK_{a1} **3.02** · pK_{a2} 4.38

Both matter: maleic wins K_{a1} (H-bond stabilises mono-anion), **loses** K_{a2}

★ BASIC STRENGTH

NH_2^- > OR^- > OH^- > ArO^- > $RCOO^-$ > H_2O | piperidine > NH_3 > pyridine > aniline > Ph_2NH > pyrrole

lower pK_b = stronger base = higher pK_a of its conjugate acid · **resonance kills basicity** · **sp³ > sp² > sp**

★ AMINES

in WATER : 2° > 1° > 3° > NH_3 (methyl) | 2° > 3° > 1° > NH_3 (ethyl) | GAS PHASE : 3° > 2° > 1° > NH_3

in water +I (favours 3°), solvation of R_nNH^+ (favours 1°) and sterics (hurt 3°) compete — the compromise puts **2° on top**

9 FACTORS THAT DECIDE BASE STRENGTH

A base is strong when its lone pair is available and its conjugate acid is stable

- **+I** ↑ basicity · **−I** ↓ basicity
- **Resonance kills basicity** (aniline, amide, pyrrole)
- **sp³ > sp² > sp**
- **Steric hindrance** at N ↓ basicity
- **Solvation** — more N–H ⇒ ↑ basicity in water
- **Aromaticity** overrides everything

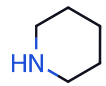
Lower pK_b = stronger base = higher pK_a of its conjugate acid

THE ALKYLAMINE PUZZLE — TWO DIFFERENT ANSWERS

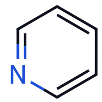
Series	Order
Gas phase (pure +I)	**3° > 2° > 1° > NH_3**
Water — methyl	$(CH_3)_2NH$ **3.29** > CH_3NH_2 **3.36** > $(CH_3)_3N$ **4.26** > NH_3 **4.75**
Water — ethyl	$(C_2H_5)_2NH$ **3.02** > $(C_2H_5)_3N$ **3.25** > $C_2H_5NH_2$ **3.29** > NH_3 **4.75**

Why the flip? +I (favours 3°) vs solvation (favours 1°) vs sterics (hurts 3°) ⇒ **2° on top** in water

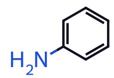
AROMATIC & HETEROCYCLIC BASES (pK b)



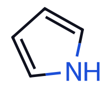
piperidine
2.8
sp³



pyridine
8.8
sp², lp in plane



aniline
9.38
lp in resonance

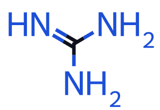


pyrrole
13.6
lp IS the aromatic sextet

Substituted anilines: *p*-toluidine 8.92 > aniline 9.38 > *p*-chloroaniline 9.85 > *p*-nitroaniline 13.0

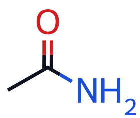
Pyrrole is NOT basic — N-protonation would kill the 6π system, so it protonates at C-2

GUANIDINE — THE STRONGEST NEUTRAL ORGANIC BASE



pK_b ≈ **0.4** · the **guanidinium ion** shares ⊕ equally over **three N** (Y-delocalisation)

WHY AMIDES ARE NOT BASIC



N lp → C=O, so N is **sp²** and the amide is **10¹⁰× less basic** than an amine. It protonates at **O**, not N.

SPOT-CHECK ORDERS

- NH_3 > PH_3 > AsH_3 (down group 15, size ↑)
- R_3N (sp³) > pyridine (sp²) > R–C≡N (sp)
- RO^- > HO^- > $RCOO^-$ > H_2O (charge + resonance)
- $NaNH_2$ > NaOH > $NaHCO_3$ (read pK_a ladder backwards)

LEAVING GROUP $N_2 > TfO^- > I^- > Br^- > TsO^- \approx MsO^- > Cl^- > H_2O > F^- \gg OH^-, OR^-, NH_2^-$

weaker base ⟹ better leaving group · activate a bad LG : protonate OH ⟶ H_2O^+, or make the tosylate / mesylate ($SOCl_2$, PBr_3)

10 ELECTROPHILE (E^+) — electron-DEFICIENT, attacks e^--rich sites

Type	Examples
Cations	H^+, NO_2^+, NO^+, Br^+, Cl^+, R^+, RCO^+, ArN_2^+
Vacant orbital (neutral)	BF_3, $AlCl_3$, $FeCl_3$, $ZnCl_2$, SO_3, $:CCl_2$
Polar multiple bond	C of **C=O**, C of **C≡N**, RCOCl, epoxide C

All **Lewis acids** are electrophiles · δ^+ carbon counts too

NUCLEOPHILE (Nu^-) — electron-RICH, donates a pair

Nucleophilicity ≠ basicity — kinetic (attack at C) vs thermodynamic (affinity for H^+)

Comparison	Order
Same atom, charge	$RO^- > ROH$ · $NH_2^- > NH_3$
Same atom, basicity	$RO^- > HO^- > RCOO^-$
Halides · protic solvent (H_2O, ROH)	**$I^- > Br^- > Cl^- > F^-$** small F^- is caged by H-bonds
Halides · aprotic solvent (DMSO, DMF, acetone)	**$F^- > Cl^- > Br^- > I^-$** “naked” anion — basicity rules
Steric bulk	*t*-BuO^- : strong **base**, poor **nucleophile**

Ambident (two attacking sites): CN^-/NC^- · NO_2^-/ONO^- · enolate (C or O) · SCN^-

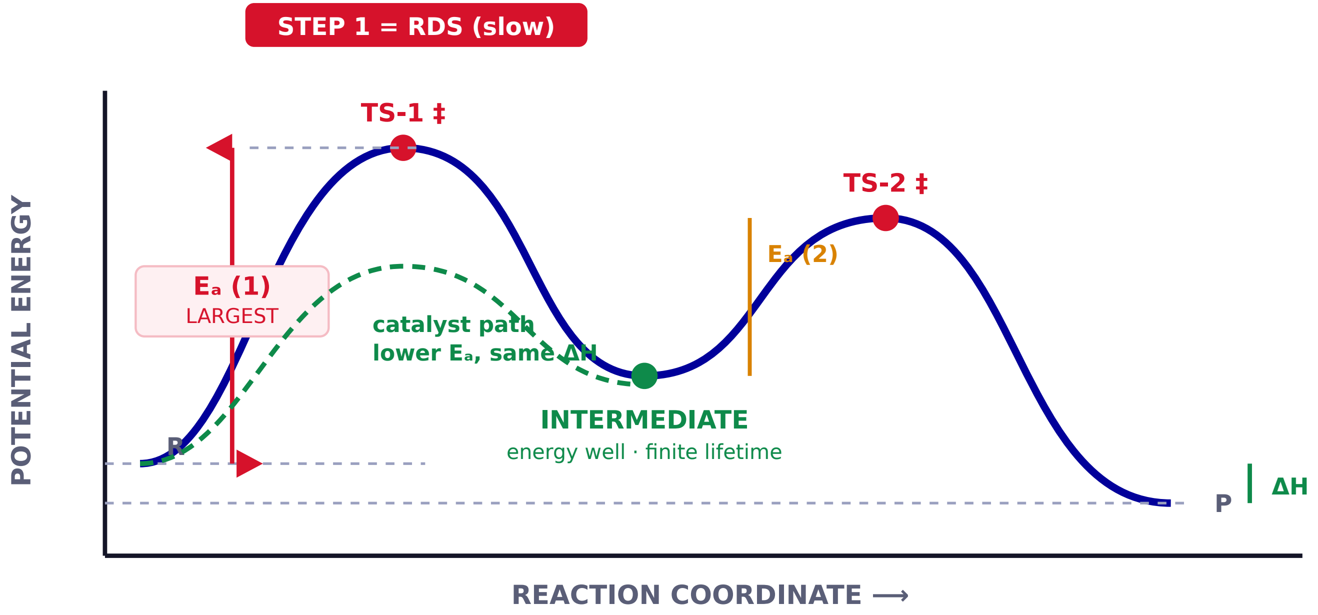
LEAVING GROUP (LG) — must depart with the pair

WEAKER BASE ⟹ BETTER LG

Rating	Group	pK_a of conj. acid
Superb	**N_2**, TfO^-	— / −14
Very good	I^-, Br^-	−10, −9
Good	TsO^-, MsO^-, Cl^-	−2.8, −2, −7
Fair	H_2O	−1.7
Poor	F^-	3.2
Never	**OH^-, OR^-, NH_2^-, H^-, R^-**	16 – 50

Fix a bad LG: protonate OH → H_2O^+, or make the tosylate / mesylate (or use $SOCl_2$, PBr_3)

11 TWO-STEP REACTION — READ EVERY FEATURE OFF THE CURVE



TRANSITION STATE vs INTERMEDIATE

	Transition state ‡	Intermediate
Position	energy **maximum**	energy **minimum** (well)
Bonds	partial, **dotted**	**full** bonds
Lifetime	≈ 10^{-13} s	finite, longer
Isolable?	**NEVER**	**sometimes**
Examples	S_N2 5-coordinate C	carbocation, radical, arenium ion

RATE-DETERMINING STEP & ARRHENIUS

RDS = slowest step = highest E_a = highest TS

Rate law = every species up to **and including** the RDS

$$k = A \cdot e^{-E_a/RT}$$

- **Catalyst** lowers E_a only — ΔH **unchanged**
- ↑T favours the larger-E_a direction

HAMMOND'S POSTULATE — “the TS resembles whichever species it is closer to in energy”

EXOTHERMIC step
Early, reactant-like TS ⇒ poor selectivity (Cl• halogenation)

ENDOTHERMIC step
Late, product-like TS ⇒ high selectivity (Br• halogenation, S_N1)

KINETIC vs THERMODYNAMIC CONTROL

	Kinetic product	Thermodynamic product
Formed	**faster** (lower E_a)	**more stable** (lower G)
Conditions	**low T**, short time, irreversible	**high T**, long time, reversible
Butadiene + HBr	**1,2-adduct** (−80 °C)	**1,4-adduct** (40 °C)

12 THE FOUR COMPETING PATHWAYS OF AN ALKYL HALIDE

S_N2 · 1 step · backside · INVERSION

$Nu^- \rightarrow C \cdots LG \rightarrow Nu—C + LG^-$

5-coordinate TS ‡ · CH_3>1°>2°>3° rate = k[RX][Nu]

E2 · 1 step · ANTI-PERIPLANAR (H & LG at 180°)

3° > 2° > 1° · Zaitsev · bulky base → Hofmann

S_N1 · 2 steps · RACEMISATION

$R_3C—LG \xrightarrow{\text{SLOW · RDS}} R_3C^+ + LG^- \xrightarrow{\text{FAST · Nu}^-} R_3C\text{–}Nu$

3°>2°>1°>CH_3 · rate = k[RX] · polar protic

REARRANGEMENT possible

E1 · 2 steps · same cation as S_N1 (they COMPETE)

$R_3C—LG \xrightarrow{\text{SLOW · RDS}} R_3C^+ \xrightarrow{-H^+\text{ (base)}} C{=}C$

High T + weak base favour E1 · Zaitsev major

DECIDING BETWEEN THEM

Substrate	Weak base/Nu (H_2O, ROH, I^-)	Strong Nu, weak base (CN^-, N_3^-)	Strong bulky base (*t*-BuO⁻, DBU)
CH_3–X	no reaction	S_N2	S_N2
1°	no reaction	S_N2	E2
2°	S_N1 / E1	S_N2	E2
3°	S_N1 / E1	E2	E2

Heat favours elimination · protic → S_N1/E1 · aprotic → S_N2

REGIO- & STEREO-CHEMISTRY OF ELIMINATION

- **Zaitsev: more** substituted alkene major — normal
- **Hofmann: less** substituted major with a bulky base (*t*-BuO⁻) or bulky/⊕ LG (−N^+R_3)
- **E2:** H and LG **anti-periplanar** · cyclohexane ⇒ **trans-diaxial**
- **E1cb:** carbanion · acidic β-H + poor LG

ADDITION REACTIONS

Type	Substrate	Attacks first
A_E electrophilic	C=C, C≡C	E^+ (H^+, Br^+)
A_N nucleophilic	C=O, C≡N	Nu^- (CN^-, RMgX)
A_R radical	C=C + HBr/peroxide	Br•

Markovnikov: H^+ adds to give the **more stable cation** · anti-M only with HBr + peroxide or hydroboration

Arenes prefer **substitution** — addition would cost 150 kJ mol⁻¹ of RE

13 OXIDATION STATE OF CARBON — THE ONLY DEFINITION YOU NEED

OXIDATION — gain O / lose H · [O], $KMnO_4$, $K_2Cr_2O_7$, PCC, CrO_3 →

CH_4	CH_3OH	HCHO	HCOOH	CO_2
alkane −4	1° alcohol −2	aldehyde 0	acid +2	carbonate +4

← **REDUCTION — gain H / lose O · [H], $LiAlH_4$, $NaBH_4$, H_2/Pd, Zn-Hg/HCl**

NAMED REARRANGEMENTS — QUICK TABLE

Name	Substrate → Product	Migration
Hofmann bromamide	$RCONH_2 + Br_2/NaOH \rightarrow RNH_2$ **(1 C less)**	C → N
Curtius	$RCON_3$ —Δ→ RNCO → RNH_2	C → N
Lossen	RCONHOH → RNH_2	C → N
Schmidt	RCOOH + HN_3 → RNH_2	C → N
Beckmann	ketoxime + H^+ → amide	**anti** group → N
Baeyer–Villiger	ketone + RCO_3H → ester	C → O
Pinacol-pinacolone	1,2-diol + H^+ → ketone	C → C^+
Wagner-Meerwein	carbocation → isomeric cation	C → C^+
Benzilic acid	benzil + OH^- → benzilic acid	C → C
Fries	aryl ester + $AlCl_3$ → *o/p*-hydroxy ketone	acyl → ring
Claisen / Cope	allyl aryl ether / 1,5-diene —Δ→	**[3,3]** sigmatropic
Favorskii	α-halo ketone + base → ester/acid	cyclopropanone

OXIDISING AGENTS

Reagent	Does
PCC / PDC / Collins	1° ROH → **aldehyde** (stops there)
$KMnO_4$ (hot, conc.)	1° ROH → acid; alkyl benzene → **PhCOOH**
$K_2Cr_2O_7/H^+$	1° ROH → acid; 2° ROH → ketone
Cold dil. $KMnO_4$	alkene → *syn*-diol (Baeyer test)
O_3 then Zn/H_2O	ozonolysis → carbonyls
Tollens / Fehling	aldehyde only → acid

3° alcohols resist mild oxidation — **no α-H**

REDUCING AGENTS

Reagent	Reduces
$NaBH_4$	aldehyde, ketone, RCOCl **NOT** ester / acid / amide / nitrile
$LiAlH_4$	**everything** · acid, ester → 1° ROH · amide, nitrile → amine
DIBAL-H (−78 °C)	ester / nitrile → **aldehyde**
Rosenmund H_2/Pd-$BaSO_4$	RCOCl → RCHO
Clemmensen Zn-Hg/HCl	C=O → CH_2 (**acidic**)
Wolff–Kishner N_2H_4/KOH	C=O → CH_2 (**basic**)
Lindlar / Na–NH_3(l)	alkyne → *cis* / *trans* alkene